

Resilience Assessment of Urban Power Network Considering Multi-Stage Electrical-Water Fault Propagation

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Abstract—Multi-voltage-level urban power network (UPN) is the basis for other infrastructure networks, especially water distribution network (WDN). Compared with other networks, UPN is more vulnerable to extreme events, including extreme weather events and deliberate attacks. The affection of UPN fault on WDN displays complex multi-stage characteristics. This paper first proposes a comprehensive tri-stage model of electrical-water fault propagation, including the electrical-water network connection model, the reduced-order hydraulic network model, and the water supply model of customers. Second, a scenario selection method and two resilience assessment indices are proposed. The indices measure the resistive capability of UPN considering the fault propagation effect. Finally, the proposed model and assessment method are verified by a simplified real-world urban power and water network based on the selected typical fault scenarios. The resilience assessment method helps electrical utilities to identify critical components of UPN and to make optimal decisions for long-term component hardening and short-term post-disaster restoration.

Index Terms—Resilience assessment, urban power network, water distribution network, fault propagation, extreme event.

NOMENCLATURE

A. Notation Associated With Electric Power Network

Parameters and Sets

i, j, Ω_{EN}	Index and set of nodes in the high-voltage system.
$\Omega_{ENF} \subset \Omega_{EN}$	Set of faulted substations.
$(i, j), \Omega_{EB}$	Index and set of branches in high-voltage system.

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$\Omega_{EBF} \subset \Omega_{EB}$	Set of faulted high-voltage branches.
h, Ω_{ENi}	Index and set of load nodes of the i -th substation.
s, S	Index and set of fault scenarios.
B_{ij}	Admittance of branch (i, j) .
$z_{ij,t}$	Switching state of branch (i, j) at t , $z_{ij,t} = 1$ if closed.
$P_{i,t}^G$	Power output of generator at node i at t .
$P_{i,t}^D$	Load demand at node i at t .
$\Delta P_{i,t}^D$	Load reduction at node i at t .
P_{ci}^D	Average load demand per customer at node i .
x_i^e, X^S	Binary parameter and set of substation state, $x_i^e = 1$ if the substation is in operation, 0 if it is faulted.
v_w	Wind speed of a hurricane.
w_d	Flood depth in the substation.
b_{ij}	Admittance of branch (i, j) .
P_{i-h}	Power demand of node h .
L_h	Distance of medium-voltage line from the substation to node h .

Variables

$P_{ij,t}$	Real power flow of branch (i, j) at time t .
ρ^S	Probability of substation transformer fault.
ρ^{MVL}	Probability of medium-voltage distribution line fault.

B. Notation Associated With Water Network

Parameters and Sets

i, j, Ω_{WN}	Index and set of nodes in water network.
$(i, j), \Omega_{WB}$	Index and set of branches in water network.
Ω_{WP}	Set of branches with water pump.
$\Omega_{WR}(l)$	Set of branches in the l loop of the water network.
$\pi(i) / c(i)$	Set of parent/children nodes of node i .
x_i^P, X^P	Binary parameter and set of water pump states, $x_i^P = 1$ if the pump is in operation, 0 if it quits operation.
W_i^T, W^T	Water plant output of node i and set of W_i^T .
W_i^D, W^D	Water demand of node i and set of W_i^D .
I	Identical matrix.
y_i, Y	Ground elevation at node i and set of y_i .
H_{ij}^P, H^P	Water head gain of pump (i, j) and set of H_{ij}^P .

H^{st}	Required urban service water head.
l_{ij}	Length of water branch (i, j).
C_{ij}	Equivalent frictional coefficient of branch (i, j).
β_{ij}	Set of water demand node along branch (i, j).
LP_l	Set of branches in the l -th loop of the WDN.
ΔH^F	Average water head loss per floor.
H^{st} / H^{1F}	Standard service water head for the 6 th /1 st floor.
<i>Variables</i>	
$H_i, \mathbf{H}$	Service (free) water head of node i and set of H_i .
$\Delta H_{ij}, \Delta \mathbf{H}$	Water head loss of branch (i, j) and set of ΔH_{ij} .
W_{ij}	Water flow from i to j (m ³ /s).
$NC_{i,t} / \Delta NC_{i,t}$	Number of all/affected customers at node i .

I. INTRODUCTION

City is the political, economic, science and technology, and cultural center of a region and even a country. The infrastructural networks, such as electrical power, water, natural gas, and communication, are the basis for the normal operation of the city. In particular, power system failure not only affects different types of customers but also causes malfunction of other networks [1], [2]. The concept of "power system resilience" is introduced to assess its ability to withstand and recover from significant damages caused by natural disasters (e.g., hurricanes, floods) or deliberate attacks [3], [4], [5].

With global climate change, extreme weather events tend to be more frequent than before [6], [7]. One type is hurricane or torrential rain that occurs in summer. For example, the Zhengzhou 7.20 torrential rain in 2021, China, destroyed over 500 overhead lines. Water treatment plants and pump stations stopped operation due to power outages [8]. Over 1.2 million customers lost power and/or water supply for at least one day. The second is snow and ice coating that occurs in winter, such as the Texas snowstorm in February 2021, the United States. Over 3.6 million customers lost power supply in this event. The power outage also led to water outages in some regions [1], [9]. Although water distribution pipelines are hardly damaged in those disaster events, severe power outages lead to a cascading effect in the water distribution network (WDN), which contains a lot of electrical-driven components [10]. Eventually, the out-of-service of electricity and water in residential, commercial, and public-service customers cause a wide social pandemic [11]. Therefore, the resilience assessment of the urban power network (UPN) should consider electrical-water fault propagation, which serves as a guideline for utility companies to efficiently recover both electricity and water supply after extreme events.

The vulnerable component identification is an essential measure of the "withstanding" aspect of the system's resilience. It means the percentage of system function loss that is caused by one or several component failures [12]. From the perspective of complex network theory, a critical node or region of the WDN can be identified by the weighted sum of its structural importance and functional importance [13], [14]. Reference [15] proposed a vulnerability assessment method of industrial WDN using

the dynamic inoperability input-output model. Overall, those works were focused on fault propagation within the WDN by specifying a faulted branch or faulted node. In disaster scenarios, however, the large-scale water outage is not caused by damage to water pipelines, but by a decrease in service water pressure due to water pump outages [16]. Therefore, the resilience assessment should be established on a hydraulic model of the WDN.

The operational interdependency of electrical-water coupled networks was addressed by many studies. The first is about the optimal energy flow [17], [18], [19] and cooperative planning of an electrical-water coupled network in normal operating condition [20], [21], [22], [23]. The purpose is to minimize the operating cost while serving the electrical and water load. Reference [21] proposed a mixed-integer nonlinear model of water-energy nexus with the objective of minimizing the operating cost of water pumps and electrical generators. The study in [22], [23] co-optimized usual electric loads and water pump loads considering the baseline demand of electricity and water customers. An optimization model was developed in [24] to promote the economic operation of coupled electrical-water networks considering the water flow dynamics. The second problem is on the post-disaster restoration of power distribution system considering electrical-water interdependency. Reference [10] proposed a restoration strategy of power distribution networks by dispatching the power consumption of water pumps and gas compressors. The study in [25] presents metrics of infrastructural-operational resilience (IOR) for real-time operations of WDN and power distribution system under extreme conditions of serious drought or electric grid failure. Based on the network interdependency, a set of metrics was introduced to measure the sensitivity of water load loss regarding electricity loss [12]. In addition to the single-directional fault propagation, the study in [26] proposed a unit commitment method of thermoelectric power plants considering the fuel and water cost under drought events. However, the existing models in [10], [12], [25], [26] cannot represent the practical fault propagation mechanism because they simplify the WDN to be a radial structure or ignore the multi-voltage-level characteristics of UPN. To our best knowledge, the urban-level electrical-water fault propagation has not been modeled in existing literatures.

Considering the fault propagation from UPN to WDN, this paper proposes a resilience assessment method of UPN. The method identifies vulnerable components in UPN according to its cascading effects. The innovations are as follows.

- A tri-stage fault propagation model is proposed to accurately quantify the impact of the UPN fault on the transmission function of the WDN. The stages include the electrical-water network coupling model, the hydraulic computation model, and the water supply model. Considering the huge number of nodes, a dimension reduction method of WDN is proposed to simplify its topological and energy flow models of the WDN for hydraulic computation.
- Based on the fault propagation model, the paper introduces two UPN resilience metrics in terms of function curve and sensitivity. Considering customers' usual demand for electricity and water supply, the quantity of electrical/water

TABLE I
COMPARISON BETWEEN DIFFERENT ENERGY NETWORK

Item	Sub-item	Electrical network	Water network
Topology feature	Minimal node	Service transformer	Water demand port, reservoir
	Branch	Distribution line	Water pipeline and water pump
	Network classification	High-voltage transmission (HVT): meshed High-voltage distribution (HVD): meshed or radial Medium-voltage distribution (MVD): radial	Main pipeline: mostly meshed Branch pipeline: mostly radial
Physical feature	Energy flow	Power flow	Water flow (incompressible fluid)
	Pressure loss	Voltage drop (caused by line impedance)	Service water head loss (caused by pipeline friction, flow direction change, and height gain)
	Theory	Ohm Law, Kirchhoff Law	Darcy-Weisbach Equation
Fragility		- The line is likely to be damaged; - The end user is totally unserved if disconnected.	- The line is unlikely to be physically damaged; - End users are partially unserved if the water head is low.

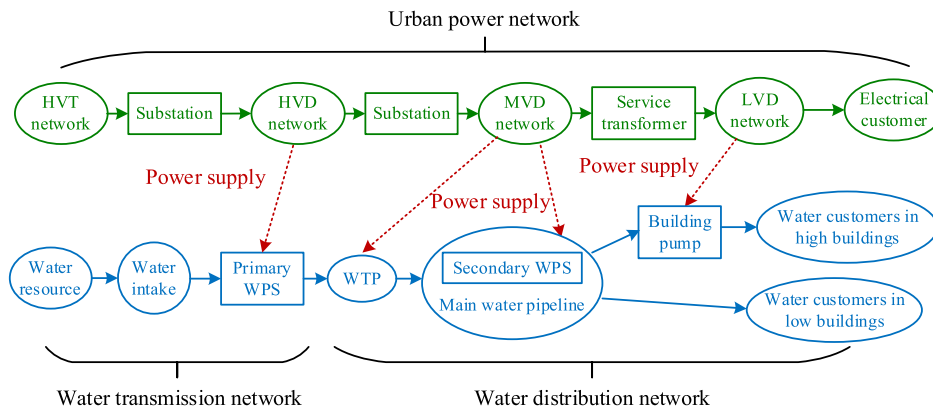


Fig. 1. Illustration of urban electrical-water interdependent network.

shortage is normalized in terms of equivalent customers. Therefore, the metrics describe the hierarchical correlation between the UPN disturbance and the affected end users and help decision-makers to allocate the emergency resources for a more efficient post-disaster restoration.

The remaining sections of this paper are organized as follows. Section II describes the characteristics of urban electrical-water networks in topological and physical aspects. Section III proposes the tri-stage model for the quantification of the electrical-water fault propagation principle. Section IV proposes the scenario generation algorithm and resilience assessment metrics of UPN considering the electrical-water interdependency. The case studies are presented in V. Section VI concludes the whole paper and introduces future research directions.

II. OVERVIEW OF ELECTRICAL-WATER NETWORK INTERDEPENDENCY

THE UPN and WDN are both energy flow networks with topology characteristics and physical characteristics. The comparison between their operating principles is the basis for fault propagation modeling, as summarized in Table I.

A. Network Topology

Fig. 1 illustrates the general coupling relation between the UPN and WDN. The red dashed arrows indicate that the operation of WDN components (e.g., pump station, water treatment

plant) depend on the power supply from UPN. Two networks are overviewed as follows.

- *Power transmission/distribution network:* The UPN is a local transmission network that consists of multiple voltage levels. High-voltage transmission (HVT) network (e.g., 220 kV in China, 115/138/230 kV in U.S.) constitutes the backbone of the network, which receives the external power injection from hundreds of kilometers away. The high-voltage distribution (HVD) network (e.g., 110 kV in China, 69 kV in U.S.) distributes the power within a region of the city. The medium-voltage distribution (MVD) network (e.g., 10 kV in China, 13/26 kV in U.S.) directly connects to the service transformer of a building or community.
- *Water distribution network:* The diameter of water pipelines ranges from 50 mm to 2000 mm [16]. The electrical-water coupling component includes water treatment plant (WTP) and water pump station (WPS) that are directly supplied by MVD system [16]. The WTP consumes electricity for purifying the water, while the WPS maintains a sufficient water head. Also, high buildings are installed with multi-level pumps for the water supply. Although 88% of the treatment facilities are equipped with backup generators, their fuel storage can only sustain for 1-2 hours, which is much shorter than the time for post-disaster power system restoration.

Due to a huge number of nodes in each network, it is not feasible to model each physical node of the network. This paper

considers the large-scale community as the minimal load node in both UPN and WDN.

B. Electrical-Water Fault Propagation

The UPN is more vulnerable to extreme weather events (e.g., hurricane, ice coating, torrential rain) because overhead power lines are more likely to be damaged than underground water pipelines [27]. Consequently, the WDN operation is affected by the power outage. On the other hand, although the thermal power plants consume a pure water, they build the water purify equipment or supplied by special pipeline instead of absorbing water from the WDN [28]. Therefore, it is reasonable to study the single-direction electrical-water fault propagation for resilience assessment of UPN with the assumption that the WDN is not damaged by extreme weather.

III. MODELING OF ELECTRICAL-WATER FAULT PROPAGATION

This section proposes a tri-stage model to describe the electrical-water fault propagation principle.

A. Overview of Electrical-Water Cascading Fault

The UPN fault affects water customers in three stages. First, the damage in the substation or overhead lines leads to the outage of coupling elements (WPS and pumps in building). Second, the absence of a water pump causes a decline in nodal water head. Finally, customers in buildings lose their water supply. Hence, the tri-stage electrical-water fault propagation model can be expressed by three independent functions.

$$\mathbf{X}^P = f_{F2P}(\mathbf{X}^S) \quad (1)$$

$$\mathbf{H} = f_{P2H}(\mathbf{X}^P) \quad (2)$$

$$\mathbf{W}_D = f_{H2D}(\mathbf{X}^S, \mathbf{H}) \quad (3)$$

In this study, the minimal node in UPN is HVT or HVD substation. Therefore, the failure state $\mathbf{X}^S$ indicates that either the substation or the distribution line connecting the substation and WPS is faulted. Functions f_{F2P} , f_{P2H} , and f_{H2D} will be studied in Section III-B–III-D, respectively.

B. Modeling of Urban Power Network in Fault Condition

1) *System Operation Principle*: This section proposes a model to characterize the impact of calculate the N-k fault on the load loss and the operation of pumps (function f_{F2P}). Generally, an UPN is partially served by the internal thermal power plant and partially by an external transmission system. The load loss is determined by three factors, including the economic dispatch of thermal power plants, network reconfiguration, and optimal load shedding [29]. The modeling of UPN consists of meshed HVT networks, substations, and HVD networks. Based on the input parameters of line fault states (z_{ij}), the simplified economic dispatch model is given by (4)–(12).

$$\min. \sum_{i \in \Omega_{EN}} (c_i^G P_{i,t}^G + c_i^D \Delta P_{i,t}^D), \quad \forall t \quad (4)$$

$$0 \leq P_{i,t}^G \leq \bar{P}_i^G, \quad i \in \Omega_{EN} \quad (5)$$

$$0 \leq \Delta P_{j,t}^D \leq P_j^D, \quad i \in \Omega_{EN} \quad (6)$$

$$\theta_{ref} = 0 \quad (7)$$

$$\sum_{j=1}^N b_{ij} \theta_j = P_{i,t}^G - (P_{i,t}^D - \Delta P_{i,t}^D), \quad i \in \Omega_{EN} \quad (8)$$

$$-M(1-z_{ij}) \leq P_{ij} - b_{ij}(\theta_i - \theta_j) \leq M(1-z_{ij}), \quad ij \in \Omega_{EB} \quad (9)$$

$$-P_{ij}^{\max} z_{ij,t} \leq P_{ij,t} \leq P_{ij}^{\max} z_{ij,t}, \quad \forall ij \in \Omega_{EB} \quad (10)$$

$$z_{ij} = 0, \quad ij \in \Omega_{EBF} \quad (11)$$

$$\sum_{ij \in \Omega_{EBLk}} z_{ij} \leq z_k^N, \quad ij \in \Omega_{EB} \quad (12)$$

The objective function (4) minimizes the total cost of generators and load shedding. Constraint (5) enforces the output limit of generator units. The load shedding limit is given by (6). The DC power flow equation is adopted to model the power flow under N-k fault scenario, as given by (7)–(10). Equation (7) specify the voltage angle of the reference bus. Constraint (8) represents the bus power balance. Note: only a small portion of nodes are installed with generator units; If there is no generator, $P_{Gj,t}$ is set to be 0. Equation (9) represents the branch power flow, which is relaxed if the branch is open. Equation (10) restricts the upper limit of the branch power flow. If branch (i, j) is faulted, then $P_{ij,t} = 0$. Equation (11) restricts the status of faulted branches. Equation (12) indicates that the HVD network maintains radial structure by opening a switch in loop k . The load shedding is converted to the number of affected customers, given by (13).

$$\Delta NC_{i,t}^E = \frac{\Delta P_{i,t}^D}{P_i^{Dc}} \quad (13)$$

A substation is de-energized if it is not supplied by the upstream system. Furthermore, the substation is identified as not being able to support the WTP if it is de-energized or the local power supply is lower than the WTP load, as given by (14).

$$\mathbf{X}^S(t) = \{x_{i,t}^S | x_{i,t}^S = 0 \text{ if } P_{i,t}^L - \Delta P_{i,t}^L < P_i^{WPS}, \\ x_{i,t}^S = 1 \text{ otherwise}\} \quad (14)$$

2) *Electrical-Water Network Connection*: The function f_{F2P} depends on the connection relationship between the UPN and WPSs. The operating state of WPSs is given by (15).

$$x_m^P = \min \left(1, \sum_{i \in \Omega_{EN}} \varphi_{mi} \cdot x_i^S \right), \quad i \in \Omega_{WPS} \quad (15)$$

where $\varphi_{mi} = 1$ if there is a power supply path from substation i to the WPS m , $\varphi_{mi} = 0$ otherwise. If the WPS is served by one or two substations, then $\sum_{i \in \Omega_{EN}} \varphi_{mi} = 1$ or $\sum_{i \in \Omega_{EN}} \varphi_{mi} = 2$. The equation can be written in a matrix format.

$$\mathbf{X}^P = \min(\mathbf{I}, \Phi \mathbf{X}^S) \quad (16)$$

where Φ is the power supply incidence matrix that consists of φ_{mi} . If the WPS is connected to a single substation, it loses power supply once it is de-energized. If it is connected to double

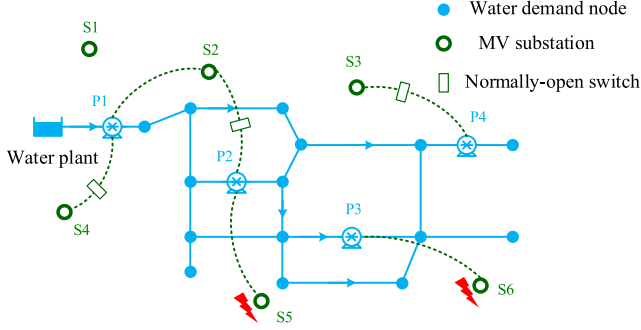


Fig. 2. Substation power supply path of the coupling element.

substations, it loses power supply only if both substations are de-energized.

We use a numerical example to illustrate the coupling relationship between substations and WPSs. As shown in Fig. 2, the system consists of four WPSs and six substations. The blue solid curve and the green dashed curve represent the water pipeline and distribution lines, respectively. Assume substations {5, 6} are faulted. The incidence matrix is

$$\Phi = \begin{bmatrix} 0 & 1 & 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 & 0 & 0 \end{bmatrix}, \mathbf{X}^S = [1, 1, 1, 1, 0, 0]^T$$

Then, the operating states of WPSs are determined.

$$\mathbf{X}^P = \min \left([1, 1, 1, 1]^T, [2, 1, 0, 1]^T \right) = [1, 1, 0, 1]^T$$

It is observed that P3 quits operation.

C. Modeling of Water Distribution Network

Function f_{P2H} is based on comprehensive hydraulic modeling of WDN. The water pump promotes the kinetic energy of water so that it can overcome the resistance of pipelines and reach end-user nodes with sufficient water head.

1) *Water Pipeline and Node*: According to the hydraulic theory, the total mechanical energy of a per-unit mass of water is called the water head (in meters). Generally, we consider the evenly-distributed, constant-speed water flow from position i to j . The Energy Equation of Water Flow describes the energy change of the per-unit mass of water from position i to j [16]:

$$y_i + H_i = y_j + H_j + \Delta H_{ij} \quad (17)$$

In a meshed WDN, the water flow depends on both the nodal water demand and the branch water head loss. According to Chézy Formula, the water head loss in the circle-intersection pipeline is expressed by (18) [16]. The absolute value sign indicates the uncertainty of flow direction in a meshed network.

$$\Delta H_{ij} = C_{ij} l_{ij}^{eq} W_{ij} |W_{ij}| \quad (18)$$

where C_{ij} is related to the diameter and material of the pipeline. Substituting (17) into (18) gives

$$H_i - H_j + y_i - y_j = C_{ij} l_{ij}^{eq} W_{ij} |W_{ij}| \quad (19)$$

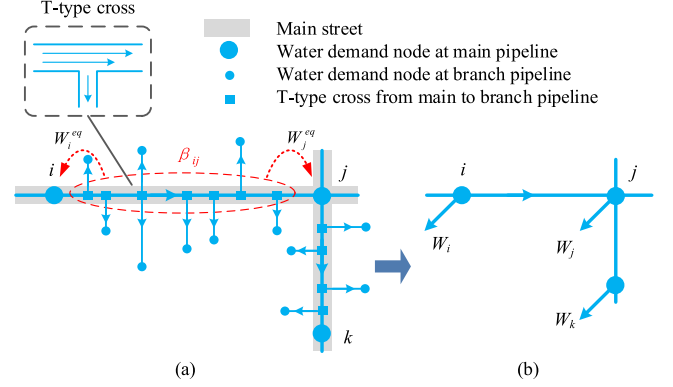


Fig. 3. Aggregation of branch WDN: (a) Real pipeline; (b) simplified pipeline.

where the equivalent length of the pipeline (l_{ij}^{eq}) combines the frictional head loss and the local head loss. The detailed method for calculating l_{ij}^{eq} is presented in Appendix. If branch (i, j) contains a WPS, the water flow equation is given by (20).

$$H_i - H_j + y_i - y_j + x_m^P \Delta H_{ij}^P = C_{ij} l_{ij}^{eq} W_{ij} |W_{ij}| \quad (20)$$

The water demand point is considered as a node in the network. The water flow balance of node j is given by (21).

$$\sum_{i \in \pi(j)} W_{ij} - \sum_{k \in c(j)} W_{jk} = W_j^T - W_j^D \quad (21)$$

For most of the nodes, $W_j^T = 0$ because there is no WTP.

2) *Network Topology Reduction*: The WDN contains a huge number of pipelines. Generally, the modeling of urban-level WDN only considers main pipelines (e.g., diameter ≥ 400 mm) and reduces the branch pipelines to water demand nodes. The “node” is identified by the cross of two main pipelines. This subsection proposes a method for WDN reduction.

- The water demand nodes along a main pipeline are aggregated and placed at the starting node and ending node as virtual demand nodes.
- Multiple T-type crosses with local water head loss can be aggregated to frictional water head loss with an equivalent length.

An example is shown in Fig. 3 to illustrate the above principle. A large number of small-diameter pipelines are connected to serve the end users. Only i and j can be regarded as nodes in the main WDN. In large-scale WDN, the total water demand along pipeline (i, j) is much smaller than its flow, that is $W_i^+ \geq \sum_{p \in \beta_{ij}} W_{p,ij}$. Thus, it is reasonable to aggregate the water demand and local head loss along the pipeline (i, j) , given by (22) [30].

$$\begin{cases} W_i^{eq} = 0.5 \sum_{p \in \beta_{ij}} W_{p,ij} \\ W_j^{eq} = 0.5 \sum_{p \in \beta_{ij}} W_{p,ij} \end{cases} \quad (22)$$

If a virtual node is connected with multiple branches (e.g., node j in Fig. 3), the total water demand of node j is obtained by

adding all virtual water demand, given by (23).

$$W_j = \sum_{l \in \pi(j) \cup c(j)} W_l^{eq} \quad (23)$$

Additionally, the details for aggregating local head loss can be found in Appendix.

3) *Electrical-Driven Pumps*: Electrical-driven water pumps are adopted to boost the water head. If the WPS operates in constant power mode, the hydraulic characteristics of a WPS is given by (24) [30].

$$\begin{cases} \Delta H_{ij}^P = H^{Pe} - \frac{\alpha}{N_p^2} W_{ij}^2 & \text{if } x_m^P = 1, m \in \Omega_{WP} \\ W_{ij} = 0 & \text{if } x_m^P = 0, m \in \Omega_{WP} \end{cases} \quad (24)$$

where H^{Pe} is the static water lift, N_p is the number of parallel pumps. Furthermore, in the variable-power mode, the power consumption of water pumps (P_{ij}^P) is a quadratic function of the water flow, given by (25).

$$P_{ij}^P = \gamma_1 W_{ij}^2 + \gamma_2 W_{ij} \quad (25)$$

where γ_1 and γ_2 are coefficients of the function.

4) *Hydraulic Computation of Meshed Network*: The hydraulic computation of meshed network is to solve nonlinear (26)–(28). Equation (26) and (27) are matrix formations of (20) and (21), respectively. Equation (28) expresses the sum of the water head in loops of the WDN.

$$\mathbf{A}^T \mathbf{H} + \mathbf{Y} + \mathbf{H}^P = \Delta \mathbf{H} \quad (26)$$

$$\mathbf{A} \mathbf{W} = \mathbf{W}^T - \mathbf{W}^D \quad (27)$$

$$\mathbf{B} \Delta \mathbf{H} = \mathbf{0} \quad (28)$$

where $\mathbf{A}$ is the $|\Omega_{WN}| \times |\Omega_{WB}|$ node-branch incidence matrix of the water network, $\mathbf{0}$ is the vector of all-zero elements. $\mathbf{A}_{ki} = 1$ if node k the starting node of branch i ; $\mathbf{A}_{ki} = -1$ if node k the ending node of branch i ; $\mathbf{A}_{ki} = 0$ if node k does not belong to branch i . $\mathbf{B}$ is the incidence matrix of the loops. Since the term $\Delta \mathbf{H}$ in (26) is a quadratic equation with an absolute sign, the hydraulic equations cannot be directly solved. The general approach is an iterative-based adjustment method, as introduced below [31].

Step 1: Specify the constant water heads of source nodes (WTPs). Also, specify the initial flow of $|\Omega_{WB}| - |\Omega_{WN}|$ branches and solve (27) for the branch flow vector $\mathbf{W}$.

Step 2: Calculate the sum of water head loss in a loop.

$$\Delta H_l = \sum_{ij \in \Omega_{Rl}} \Delta H_{ij} = \sum_{ij \in \Omega_{Rl}} C_{ij} l_{ij}^{eq} (W_{ij}^{(0)})^2 \quad (29)$$

If ΔH_l is larger than the error tolerance (e.g., 0.1m), then go to Step 3, otherwise stop.

Step 3: According to Hardy Cross Algorithm, calculate the loop corrective flow ($\Delta W_{ij}^{(0)}$) and add $\Delta W_{ij}^{(0)}$ to the initial value ($W_{ij}^{(0)}$), as given by (30) and (31).

$$\Delta W_{ij}^{(0)} = - \frac{\Delta H_{ij}^{(0)}}{\sum_{ij \in R_l} \frac{2H_{ij}^{(0)}}{C_{ij} l_{ij}^{eq}}} \quad (30)$$

$$W_{ij}^{(1)} = W_{ij}^{(0)} + \Delta W_{ij}^{(0)} \quad (31)$$

Then, return to Step 2.

Step 4: When Step 2 converges, the branch flow is also determined. Calculate the set of water head ($\mathbf{H}$) from the water plants to downstream nodes according to (26). Finally, the propagation function f_{P2H} is obtained.

D. Aggregated Modeling of Water User

A WDN node represents a community or a building. If the nodal water head falls below the required water head, customers on the high floors of the building may lose water supply. This correlation can be modeled by an explicit function f_{H2D} , which is related to the height and water supply type of buildings. This study establishes f_{H2D} by differentiating low buildings (≤ 6 floor) and high-rise buildings (> 6 floor), respectively. Two assumptions are made to simplify the model of building water supply.

- The number of customers is evenly distributed on each floor.
- The building water pumps and other electrical loads are supplied by the same service transformer. Also, there is no backup power source in the building.

1) *Low Buildings*: Generally, H_i is designed for the full water supply of low buildings without built-in pumps [32]. According to (18) and (19), the water head loss is proportional to the height of the vertical pipeline. Therefore, if $H_i < H^{st}$, then ΔNC_i is estimated by (32). Note: the index i is omitted in (32) because they correspond to the same node in the WDN.

$$\Delta NC_i = \frac{NC_i^W}{N_i^F} \cdot \max \left(N_i^F - \left\lfloor \frac{H_i - H^{1F}}{\Delta H^F} \right\rfloor, 0 \right), \quad i \in \Omega_{WN} \quad (32)$$

where $\lfloor \cdot \rfloor$ is the rounding down sign, $\Delta H^F = (H^{st} - H^{1F})/6$ represents the water head loss per floor. For example, $H_i = 23$ m, $H^{st} = 28$ m, $H^{1F} = 10$ m, $N^F = 5$, and $NC_i = 3600$. Thus, $\lfloor (23 - 10)/3 \rfloor = 4$ indicates that H_i can satisfy the water supply of 1 ~ 4 floors. Then, $\Delta NC_i = (3600/5) \cdot \max(5 - 4, 0) = 720$.

2) *High-Rise Buildings*: Since the service water head is designed to supply the 6th floor or below, the high-rise building is equipped with a building pump to boost the water to above 6th floors, as shown in Fig. 4 [32]. The affected customers of the lowest 6 floors is calculated in the same way as those of low buildings, while affected customers above the 6th floor depends on the working status of building pumps. Then, ΔNC_i can be calculated by (33).

$$\begin{aligned} \Delta NC_i = & \frac{NC_i^W}{6} \max \left(6 - \left\lfloor \frac{H_i - H^{1F}}{\Delta H^F} \right\rfloor, 0 \right) \\ & + (1 - x_i^{BP}) \frac{NC_i^W (N_i^F - 6)}{6} \end{aligned} \quad (33)$$

where x_t^{BP} is the status of the building pump, $x_t^{BP} = 1$ if the pump is in operation and 0 otherwise.

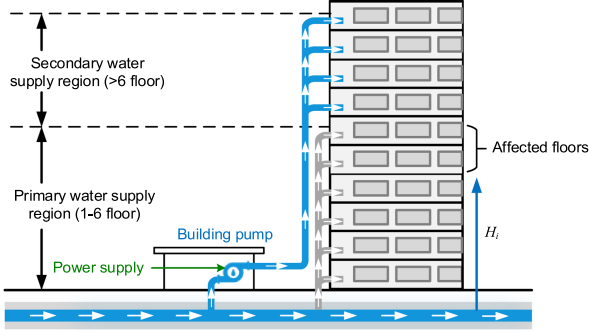


Fig. 4. Water supply in high-rise buildings.

IV. RESILIENCE ASSESSMENT FRAMEWORK

This section proposes a resilience assessment method of UPN based on the electrical-water fault propagation model. The UPN resilience is measured by its function loss and cascading effect under a sufficient number of representative fault scenarios.

A. UPN Fault Scenario Generation

The power supply path from power plants to customers includes HVT/HVD lines, MVD lines, and substations. The fault of any element may terminate the power supply path. This paper considers the fault scenarios in HVD and MVD levels in combined hurricane and rainfall weather, which is based on the empirical fragility model of components. Considering the large number of HVD substations, it is necessary to select representative fault scenarios according to their probability and impact [33].

1) *Fragility of Substations*: The substation is likely to be tripped when the transformer is inundated in strong rainfall. The fragility curve shows a substation's failure probability versus the water depth under flood scenarios. An empirical function is expressed by (34) [34].

$$\rho_i^S = \begin{cases} 0 & w_d < 0.3\text{m} \\ 0.434w_d - 0.134 & 0.3\text{m} \leq w_d < 2.6\text{m} \\ 1 & w_d \geq 2.6\text{m} \end{cases} \quad (34)$$

w_d is determined by the regional rain precipitation amount and terrain of this substation.

2) *Fragility of Overhead Lines*: The overhead line is vulnerable to strong wind. This study considers a simplified fragility model, in which the fault probability is fitted by a power function of wind speed based on the historical raw data [35]. The fault probability of an overhead line is

$$\rho^{MVL} = aL(v_w)^c \quad (35)$$

where a and c are empirical parameters, L is the line length. a and c are different in high voltage overhead lines and medium voltage ones, respectively. A set of typical parameters are $a = 2.2 \times 10^{-17}/\text{km}$ and $c = 9.91$ [35].

3) *Aggregation of MVD System*: On the preliminary that the upstream supply path of an HVD is normal, the downstream load can be supplied only if the substation transformer and the MVD line is normal (shown in Fig. 5(a)). This study proposes

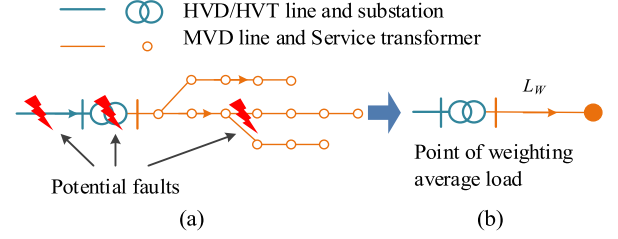


Fig. 5. Supply path of end customers: (a) Original network; (b) aggregated MVD.

a center-of-load point (CLP) via a weighting average of line distance to calculate the overall probability of line fault (shown in Fig. 5(b)). The line distance from a substation to CLP (L_{CLP}) is defined as:

$$L_{CLP} = \frac{\sum_{h \in \Omega_{ENi}} P_h L_h}{\sum_{h \in \Omega_{ENi}} P_h} \quad (36)$$

Substitute (35) into (36) gives the fault probability of the aggregated MVD network. In coastal areas, the hurricane, accompanied by strong rainfall, is likely to cause both line faults and substation faults. Therefore, the fault probability of MVD-level supply path (ρ_i^{SMV}) is

$$\rho_i^{SMV} = 1 - (1 - \rho_i^S) (1 - \rho_i^{MVL}) \quad (37)$$

4) *Disaster Scenario Generation*: Without losing generality, the disaster scenario s is defined as a combination of the set of faulted substations and the set of faulted branches, given by (38).

$$\Omega_{EFs} = \{\Omega_{ENFs}, \Omega_{EBFs}\} \quad (38)$$

The first step is to sample the faulted substation set. In an electrical-water coupling network, the impact of a substation fault is measured by the weighted sum of different types of loads within the power supply area. We introduce a three-dimensional fault feature index (ψ_i) to quantify the probability and impact of MVD power supply path failure.

$$\psi_i = (\rho_i^{SMV}, \psi_i^E, \psi_i^W) = \left(\rho_i^{SMV}, P_i^D, \sum_{i \in \Omega_{ENi}} P_i^{WP} \right) \quad (39)$$

For example, $\psi_i = (0.16, 35\text{MW}, 4.8\text{MW})$ indicates that substation i supplies 35MW electrical load and 4.8MW water pump load, while the probability of losing these loads is 0.16. If there is no WPS supplied by the substation, then $\psi_i^W = 0$. Based on the specified thresholds of ψ_i , we select critical substations by eliminating those with low failure probability or low importance.

$$\Omega_{ENFs} = \{i_1, \dots, i_{KS_s} | \psi_i^E > \underline{\psi}^E, \psi_i^W > \underline{\psi}^W,$$

$$\rho_i^{SMV} > \underline{\rho}^{SMV}, \underline{KN} \leq KN_s \leq \overline{KN}\} \quad (40)$$

where KN_s is the number of faulted substations in scenario s , $\underline{KN}$ and $\overline{KN}$ are the specified threshold according to the total number of nodes in an UPN.

The second step is to generate the faulted branch set. This study considers a N-k branch faults that occurs in heavy-loaded

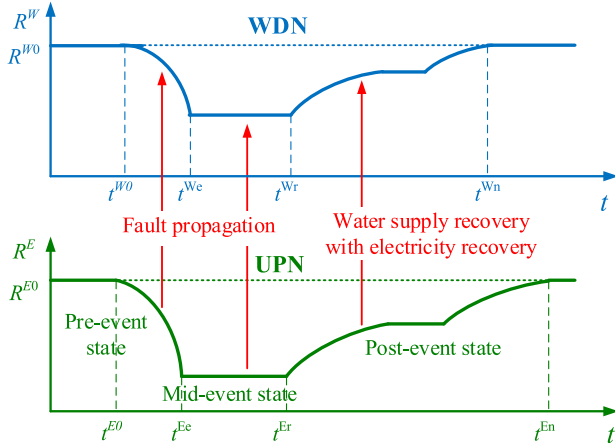


Fig. 6. Function curve of an electrical-water interdependent network.

branches.

$$\Omega_{EBFs} = \{(i_1, j_1), \dots, (i_{KB_s}, j_{KB_s}) | P_{ij,t} \geq \underline{P}_{ij}, \underline{KB} \leq KB_s \leq \overline{KB}\} \quad (41)$$

where KB_s is the number of faulted HVD branches in scenario s , $\underline{KB}$ and $\overline{KB}$ are the specified threshold according to the network dimension, $z_{ij} = 0$ if $(i, j) \in \Omega_{EBF,s}$.

Finally, the above procedure is conducted repetitively to obtain the fault scenario set that is caused by the specified intensity of an extreme weather event.

B. Resilience Assessment Metrics

The resilience assessment of UPN is to measure the impact of typical disaster scenarios on both electricity and water customers based on the fault propagation model [12]. As visualized in Fig. 6, with the progress of electrical load restoration, the WDN gradually recovers to the normal state. The unserved electrical load (in MW) and water loads (in m³/s) are not comparable. Therefore, it is reasonable to use the number of affected customers to quantify the function curve of UPN and WDN, as given by (42).

$$\begin{cases} \Delta NC_{t,s}^E = \sum_{i \in \Omega_{EN}} \Delta NC_{i,t,s}^E, & t^{E0} \leq t \leq t^{En} \\ \Delta NC_{t,s}^W = \sum_{i \in \Omega_{WN}} \Delta NC_{i,t,s}^W, & t^{W0} \leq t \leq t^{Wn} \end{cases} \quad (42)$$

Two metrics are proposed to measure the impact of UPN fault on the operation of both networks. The first one is a sequential function curve under a particular fault scenario, and the second one is a comparison between the degradations UPN and WDN under a set of typical fault scenarios.

1) *Scenario-Based Function Degradation*: After severe disasters, some customers suffer electricity outages, some suffer water outages and some suffer both. The indices $R_{s,t}^W/R_{s,t}^E$ is defined as the ratio of served electrical/water customers. They present a time-of-use function degradation of UPN/WDN under

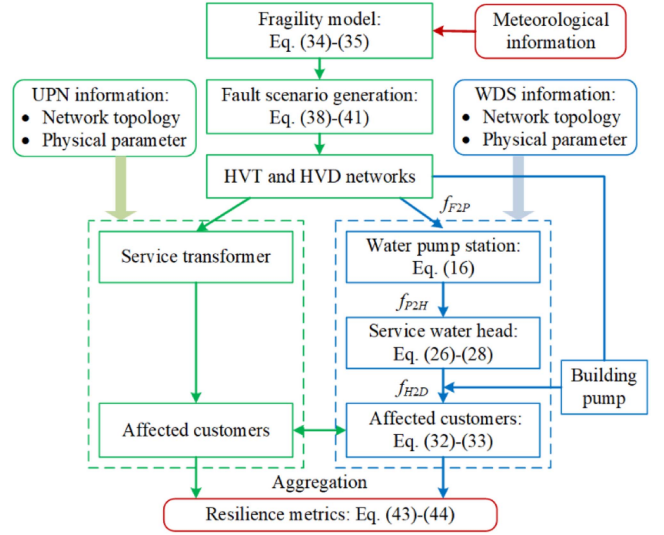


Fig. 7. UPN resilience assessment considering electrical-water fault propagation.

a particular fault scenario.

$$\begin{cases} R_{s,t}^E = 1 - \frac{\Delta NC_{t,s}^E}{NC_{t,s}^E}, & s \in S \\ R_{s,t}^W = 1 - \frac{\Delta NC_{t,s}^W}{NC_{t,s}^W}, & s \in S \end{cases} \quad (43)$$

2) *Relative Impact of UPN Disruptions on WDN Operation*: This matrix measures the sensitivity of WDN function degradation to the component fault of UPN in the mid-event state [12], which is calculated by:

$$RI_{E2W} = \frac{\sum_{s \in S} \Delta NC_{t,s}^W}{\sum_{s \in S} \Delta NC_{t,s}^E}, \quad \max(t^{Ee}, t^{We}) \leq t \leq \min(t^{Er}, t^{Wr}) \quad (44)$$

The RI_{E2W} values larger than one indicates a higher degradation on the WDN compared to the UPN, and values lower than one indicates a less dependent operation of the WDN to meet the water demand despite the UPN degradation.

Based on the proposed fault propagation model, scenarios generation method, and resilience metrics, we summarize the resilience assessment method of UPN in Fig. 7. In each scenario, the sequential simulation of an electrical-water coupled network is conducted considering the degradation and restoration of UPN. Meanwhile, the impact on WDN is estimated by the proposed tri-stage fault propagation model. First, the states of WPS and building pumps are determined. Then, the nodal water head is calculated and the number of affected customers is determined. With the progress of UPN restoration, the number of supplied water customer is also estimated. Finally, two resilience metrics are obtained and the substation vulnerability can be identified accordingly.

V. CASE STUDIES

This section presents case studies on a real-world urban infrastructure test system. The computational tasks are performed on

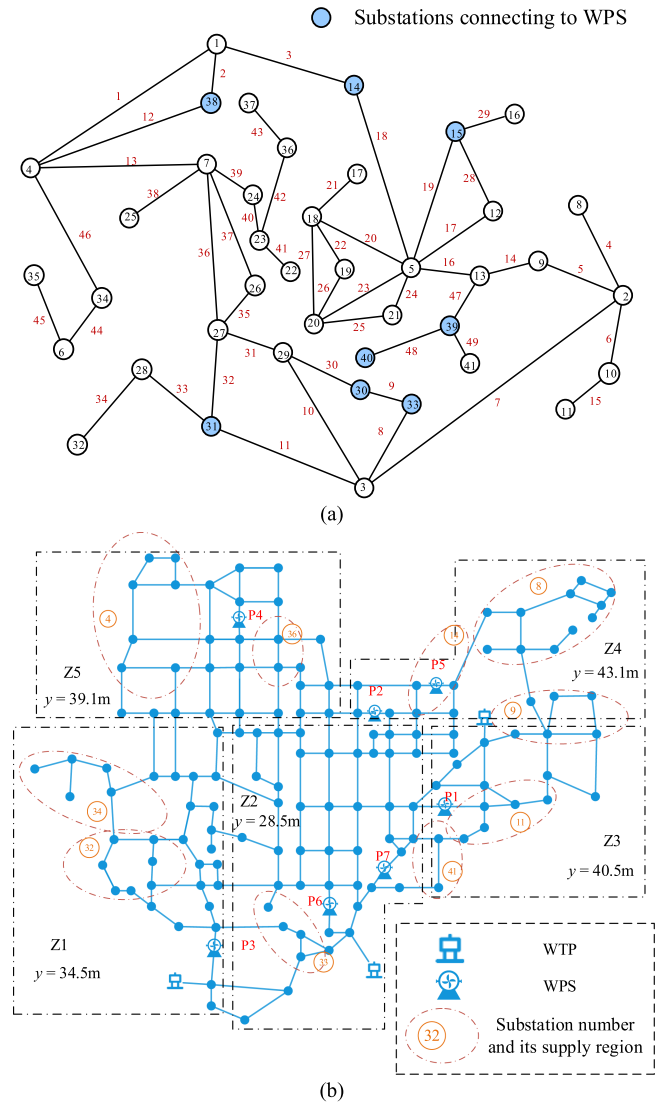


Fig. 8. Real-world urban electrical-water coupled network: (a) High voltage electrical network; (b) main water network.

a personal laptop computer with an Intel Core i7 Processor (2.20 GHz) and 16-GB RAM. The optimization model is implemented via the Matlab-based IBM ILOG CPLEX Optimization Studio V12.8.0.

A. System Description

This study considers a real-world UPN and WDN in a coastal city in China. There is a total number of 625.45 thousand customers. The approximate geographical diagrams for UPN and WDN are presented in Fig. 8(a) and (b), respectively. The UPN consists of 41 nodes (30 HVT substations and 11 HVD substations) and 50 lines with a daily peak load of 3281 MW. The WDN consists of 151 nodes, 196 branches, and 7 WPSs (labeled as P1 ~ P7) with a total daily water demand of 154.8 thousand m^3 . In the WDN, 398.0 thousand customers are located at the 1 ~ 6 floors and their water supply can be guaranteed by the service water head. The WDN is divided into five zones (labeled as Z1

TABLE II
PARAMETERS OF WDN

Pump	Coupled UPN node	Static water head lift (m)	Normal flow rate (m^3/s)	Rated power (MW)
1	39	18.0	5.74	1.35
2	15, 17	27.3	24.43	8.71
3	31	35.9	9.75	4.57
4	38	20.1	26.19	6.87
5	14	40.7	12.36	6.58
6	30, 33	32.8	14.09	6.04
7	40	37.2	13.21	6.43

TABLE III
UPN FAULT SCENARIOS

Scenario No.	Faulted branches	Faulted substation	Affected WPS
1	13, 39, 42	38	4
2	3, 5, 19, 28	31	2, 3
3	3, 18, 14, 28	15	5
4	7, 8, 31, 34	39	1
5	3, 13, 14, 47	14, 36	1, 5, 7
6	14, 18, 40, 42	14, 38	4, 5
7	3, 7, 14, 15	33, 39	1
8	7, 8, 14, 18, 31	14, 30, 33	5, 6
9	8, 14, 19, 31	30, 31	3
10	5, 13, 18, 31, 33	15, 17	2
11	7, 8, 14, 18, 42, 48	11, 15, 36	7

~ Z5) according to the average ground elevation that is marked in Fig. 8(b). The parameters of WPSs are listed in Table II, in which pumps P2 and P6 are connected to two substations and others are connected to one substation. Furthermore, there are three WTPs. Two WTPs are located at the southernmost of the city and one at the northeast part.

Considering a hurricane event in the UPN, the spatial distribution of wind speed ranges from 32 m/s to 38 m/s, and flood depth ranges from 0.4 m to 1.5 m. Then, critical UPN branches and substations are selected according to the proposed fragility models in Section IV-A. Considering $N - KB$ ($2 \leq KB \leq 5$) branch faults and $N - KN$ ($2 \leq KN \leq 4$) substation faults, we select 11 typical fault scenarios based on the proposed method, as listed in the 2nd and 3rd columns of Table III (The label of nodes and branches are shown in Fig. 8(a)). The affected WPSs are obtained by the proposed functions f_{F2P} , as listed in the 4th column. Note: the 3rd column only includes the substations that are de-energized by flood. Other substations that are isolated by branch faults are not listed. Take scenario 8 as an example, Branches {7, 8, 14, 18, 31} are faulted. Consequently, substations {2, 8, 9, 10, 11} are de-energized because they are disconnected from UPN. Meanwhile, substations {14, 30, 33} are de-energized because the main transformer or secondary feeder is faulted. Finally, the total set of de-energized substations is {2, 8, 9, 10, 11, 14, 30, 33}. According to the fault propagation model, P5 and P6 quit operation because they lose the upstream power supply.

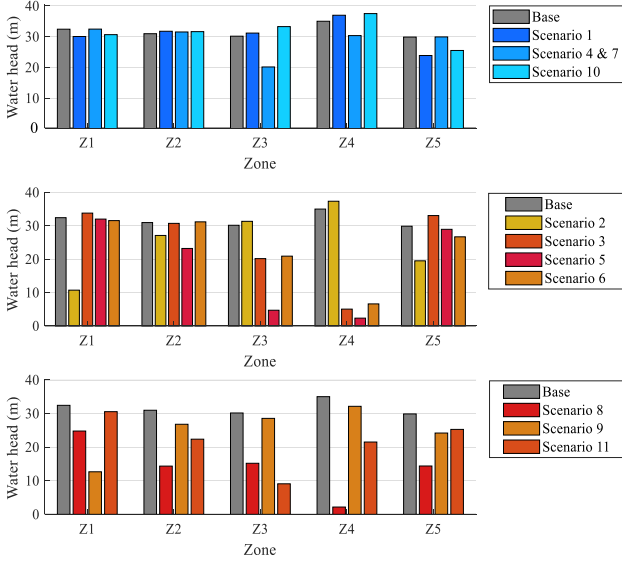


Fig. 9. Zonal average water heads in typical fault scenarios.

B. Simulation Result

1) *Water Head and Affected Customers*: Based on the status of WPSs in 11 scenarios, the nodal water head is obtained by f_{P2H} . Then, the average water head of all scenarios are presented in Fig. 9, in which the “Base” bar indicates that all WPSs normally operates. The simulation time for the tri-stage fault propagation of each scenario ranges from 0.2 to 0.3 second. Overall, it can be observed that the contribution of each WPS to the zonal average water head differs significantly. The result is analyzed as follows.

- **Scenarios with small water head decrease**: The similarity in scenarios 1, 4, 7 and 10 is that the water head only decrease a small portion in Z1 and Z2. Hence, these two zones are not sensitive to pumps P1, P2 and P4. In scenarios 1 and 10, the water head of Z3 and Z4 even increase by a little portion. This is because P2 and P4 mainly boosts the water head for Z5 and their effect contradict with other WPSs that are located at the southern part of the WDN.
- **Scenarios with large water head decrease**: In most scenarios, there is a significant decrease of water head. In particular, the similarity in scenarios 3, 5, 6 and 8 is that the average water head of Z4 decreases below 10 m and P5 quits operation. Thus, there is an obvious correlation between these two events. The significant water head decrease occurs for two reasons: 1) the high-elevation zone Z4 is a receiving end of the WDN, and 2) the water head of Z4 highly relies on P5 that locates at the main water delivery path to this zone. Therefore, in the meshed WDN, some zones are sensitive to one or two WPS due to the ground elevation and the network topology.

According to function f_{P2H} , the number of affected customers depends on both the water head and the height of the building. Take scenarios 8 as an example, the number of unserved water customers in each zone is shown in Fig. 10. We can observe that

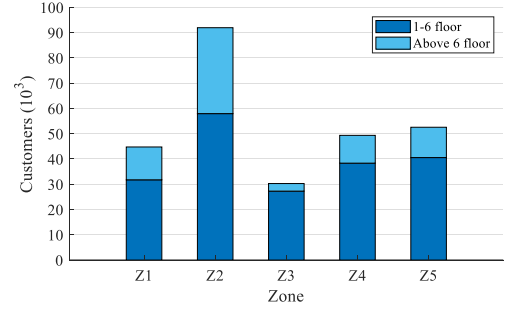


Fig. 10. Vertical distribution of unserved water customer (in scenario 8).

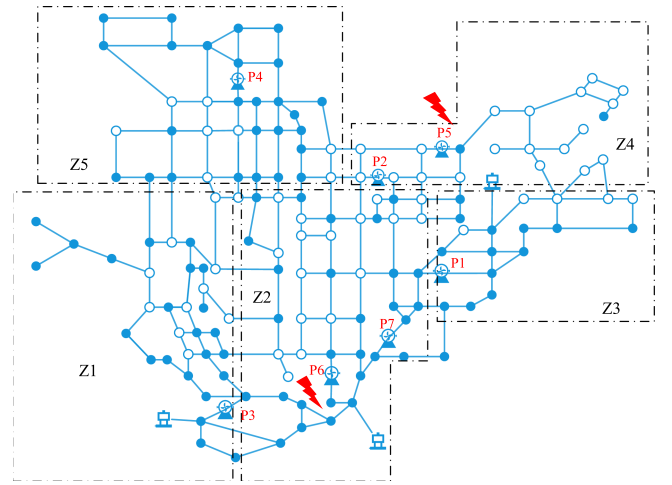


Fig. 11. Spatial distribution of affected water node (in scenario 8).

in Z4, the low water head causes a large percentage of customer to lose water supply (89%).

2) *System Function Curve*: The system function curve illustrates the coupling between UPN and WDN in multiple operating states. Take Scenario 8 as an example, three WPSs stop operation due to UPN faults. Then, the number of affected water customers is estimated by the fault propagation model. The spatial distribution is visualized in Fig. 11. In this figure, those nodes where there is more than 1000 affected customers are marked with hollow circle. At each time step, we obtain the electrical load loss by using the optimal load shedding model (in (4)–(12)). The time-of-use function curve UPN and WDN is demonstrated in Fig. 12, indicating their resistive and recovery capability under a specified UPN fault scenario.

In Scenario 8, P6 and P5 are assumed to be restored at $t = 7$ and $t = 10$, respectively. The number of served customers can be simulated based on the fault propagation model. Thus, the milestones of the system restoration are marked in the figure. When $t = 3 \sim 6$, both P5 and P6 stop operation due to power outage and only 350 out of 625 thousand customers have water supply. At $t = 7$, around 150 customers restore water supply because P6 starts operation. This result indicates a high sensitivity of P6 in the resilience curve of WDN. At $t = 10$, P5 is restored so that water heads of all nodes recover to normal value. However, the water supply of high-rise buildings is not

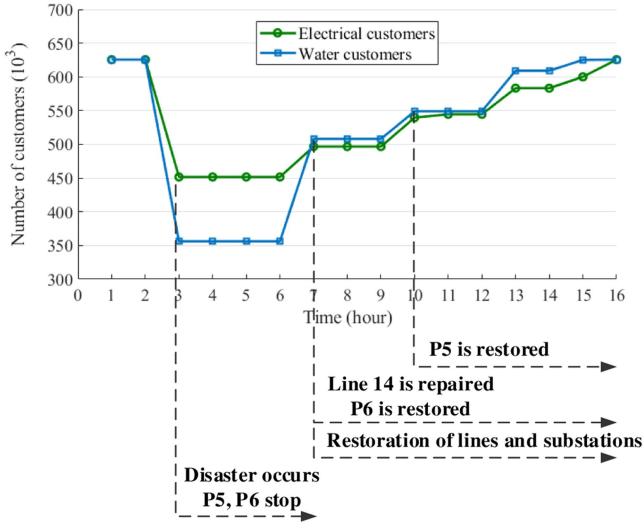


Fig. 12. Resilience curve of electrical-water network considering the fault propagation (in scenario 8).

TABLE IV
ELECTRICAL-WATER LOAD LOSS UNDER FAULT SCENARIOS

Scenario No.	$\Delta NC_{t,s}^E$ (In thousand)	$\Delta NC_{t,s}^W$ (In thousand)	RI_{E2W}
1	58.26	42.10	0.72
2	129.65	154.87	1.19
3	101.04	76.51	0.76
4	83.58	32.68	0.39
5	122.49	243.4	1.99
6	113.58	99.85	0.88
7	142.95	29.28	0.20
8	173.87	269.41	1.55
9	65.52	122.48	1.87
10	155.8	45.31	0.29
11	115.29	123.43	1.07

restored until the power supply path (including HVD and MVD branches) to this building is restored. In the restorative period ($t = 3 \sim 16$), this study adopts an empirical scheduling of UPN restoration. The restoration of WDN synchronizes with UPN even if it is not damaged under the extreme weather event. Therefore, the resilience curve in Fig. 12 suggests that the de-energized substations which supply WPSs should be restored in higher priority under the event. Furthermore, if multiple UPN branches/substations are faulted, the electric utility company should make an optimal plan of UPN restoration to enhance the resilience curves $R_{s,t}^W$ and $R_{s,t}^E$. However, the detailed technique is beyond the scope of this paper and left for future works.

3) *Relative Impact of UPN Fault on WDN*: In comparison with the resilience curve, the relative impact measures the mid-event system resilience in the scenario dimension. In this study, the mid-event period of each scenario is assumed to be $t = 4 \sim 6$ (the same as Fig. 12). The loss of electrical load and water load of 11 typical scenarios are obtained by the fault propagation model, as listed in Table IV. In some scenarios (e.g., 7 and 10), the affected water customers are much fewer than affected electrical customers for two reasons: 1) in areas with low buildings, the impact of water head decrease is not significant; 2) only a small portion of high buildings suffer

power outage. In other scenarios (e.g., 5 and 8), the loss of water customers is larger than electrical customers because the water head is seriously reduced by the power outage (shown in Fig. 9). Finally, according to (44), $RI_{E2W} = 1.018$ for the 11 typical scenarios. This index reflects the impact UPN interruption on WDN under stochastic scenarios. The reason for high RI_{E2W} is that the UPN fault causes more WPSs and building pumps to quit operation. According to the result in Table IV, RI_{E2W} in Scenarios 5 and 9 is much larger than others. Meanwhile, substation #14 is de-energized in both scenarios. Therefore, the substation 14 (supplying P5) is a critical node in UPN and branches 13 and 18 (connecting to node 14) are critical branches from the aspect of electrical-water fault propagation.

In summary, the power utilities are suggested to harden the power supply path of WPSs (P4, P5 and P6) and install more emergency generators to reduce the impact of UPN fault on WDN.

VI. CONCLUSION

This paper proposes an integrated resilience evaluation method of UPN considering its function correlation with WDN. The tri-stage fault propagation model is first proposed to accurately quantify the impact of multi-dimensional UPN fault on the horizontal and vertical distribution of affected water customers. The fault propagation model illustrates the water customers' internal dependency on UPN. This is the first contribution of the paper. The typical fault scenarios are selected according to their fragility feature and impacts. Based on the affected water load and electrical load that are normalized to be comparable, resilience metrics are proposed to quantify the time-of-use function curve of UPN and the impact of UPN fault on WDN. The case study on real-world UPN and WDN indicates the sensitivity of WDN interruption with the UPN fault. Furthermore, the study extends the concept of power system resilience considering the correlation between power system and other infrastructure networks. The resilience assessment result is an essential guidance for the resilience-oriented planning and post-extreme-event restoration of UPNs.

With the consideration of the power supply dependency of WDN, the optimal UPN restoration scheduling will be the future research direction.

APPENDIX

The water head loss consists of frictional water head loss and local water head loss.

$$\Delta H_{ij} = \Delta H_{f,ij} + \Delta H_{c,ij} \quad (45)$$

$$\Delta H_{f,ij} = \frac{\lambda l_{ij}}{D} \frac{v^2}{2g} \quad (46)$$

$$\Delta H_{c,ij} = \zeta_C \frac{v^2}{2g} \quad (47)$$

where $\Delta H_{f,ij}$ is the frictional water head loss, $\Delta H_{c,ij}$ is the local water head loss, v_i is the water flow speed of at I , λ is related to the diameter and material of the pipeline, ζ_C is an empirical coefficient [16]. To simplify the hydraulic computation, a virtual

pipeline with frictional water loss (l_c) is adopted to represent the local loss, given by (48) and (49).

$$\zeta_C \frac{v^2}{2g} = \frac{\lambda l_c}{D} \frac{v^2}{2g} \quad (48)$$

$$l_c = \zeta_C D / \lambda \quad (49)$$

where λ is the friction coefficient of the pipeline, D is the diameter of the pipeline (m), ζ_C is the local water head coefficient. Substituting (46) and (48) to (45) gives (50).

$$\Delta H_{ij} = \frac{\lambda (l_{ij} + l_c)}{D} \frac{v^2}{2g} = \frac{\lambda l_{ij}^{eq}}{D} \frac{v^2}{2g} \quad (50)$$

where the equivalent length of pipeline $l_{ij}^{eq} = l_{ij} + l_c$. In hydraulic computation, utility companies pay attention to water flow instead of water velocity. The water flow of the circle pipeline is $v = 4W/(\pi D^2)$. Thus, (50) is reformulated to (18).

$$\Delta H_{ij} = C l_{ij}^{eq} W_{ij}^2 \quad (51)$$

where the equivalent frictional coefficient (C) of the water pipeline is:

$$C = \frac{8\lambda}{\pi^2 g D^5} \quad (52)$$

$$l_{ij}^{eq} = l_{ij} + |\beta_{ij}| l_c \quad (53)$$

where β_{ij} is the set of water demand nodes along line (i, j), $\zeta_C = 0.1$ for the T-type cross pipeline [16].

REFERENCES

- [1] J. W. Busby et al., "Cascading risks: Understanding the 2021 winter blackout in Texas," *Energy Res. Social Sci.*, vol. 77, 2021, Art. no. 102106.
- [2] E. Pourmaras et al., "Cascading failures in interconnected power-to-water networks," *ACM SIGMETRICS Perform. Eval. Rev.*, vol. 47, no. 4, pp. 16–20, 2020.
- [3] President's Council of Economic Advisers & U.S. Dept. Energy's Office of Electricity & Energy Reliability, "Economic benefits of increasing electric grid resilience to weather outages," U.S. Dept. Energy, Executive Office President, Washington, DC, USA, Rep., 2013. Accessed: Apr. 18, 2025. [Online]. Available: https://www.energy.gov/sites/prod/files/2013/08/f2/Grid%20Resiliency%20Report_FINAL.pdf
- [4] "Enhancing distribution resiliency: Opportunities for applying innovative technologies," *Elect. Power Res. Inst.*, Palo Alto, CA, USA, Rep. 1026889, Jan. 2013. Accessed: Apr. 18, 2025. [Online]. Available: <https://restservice.epri.com/publicdownload/000000000001026889/0/Product>
- [5] H. Han et al., "Security constrained distributed transaction model for multiple prosumers," *CSEE J. Power Energy Syst.*, vol. 10, no. 2, pp. 834–843, Mar. 2024.
- [6] M. Huang et al., "Increasing typhoon impact and economic losses due to anthropogenic warming in Southeast China," *Sci. Rep.*, vol. 12, 2022, Art. no. 14048.
- [7] C. Lin et al., "An online resilience assessment method for islanded distribution systems considering uncertainty of intermittent RESs and loads," *Energy Rep.*, vol. 8, pp. 639–649, 2022.
- [8] Z. Tao and L. Han, "Emergency response, influence and lessons in the 2021 compound disaster in Henan Province of China," *Int. J. Environ. Resource Public Health*, vol. 19, no. 1, 2022, Art. no. 488.
- [9] R. Oxner, "Texans now face a water crisis after enduring days without power," 2021. [Online]. Available: <https://www.texastribune.org/2021/02/19/texas-water-power-outages/>
- [10] J. Li et al., "Resilience-motivated distribution system restoration considering electricity-water-gas interdependency," *IEEE Trans. Smart Grid*, vol. 12, no. 6, pp. 4799–4812, Nov. 2021.
- [11] C. Zhao, N. Li, and D. Fang, "Criticality assessment of urban interdependent lifeline systems using a biased PageRank algorithm and a multilayer weighted directed network model," *Int. J. Crit. Infrastructure Protection*, vol. 22, pp. 100–112, 2018.
- [12] L. Rodriguez-Garcia, M. M. Hosseini, T. M. Mosier, and M. Parvania, "Resilience analytics for interdependent power and water distribution systems," *IEEE Trans. Power Syst.*, vol. 37, no. 6, pp. 4244–4257, Nov. 2022.
- [13] F. Wang et al., "Emergency repair scope partition of city water distribution network: A novel approach considering the node importance," *Water Resource Manage.*, vol. 31, pp. 3779–3794, 2017.
- [14] M. Zhang et al., "Framework of identifying the critical region in water distribution network for reinforcement strategy from preparation resilience," *Sustainability*, vol. 12, no. 21, Nov. 2020, Art. no. 9247.
- [15] Y. Zhang et al., "Industrial water network vulnerability analysis using dynamic inoperability input-output model," *J. Environ. Manage.*, vol. 314, Jul. 2022, Art. no. 115015.
- [16] L. W. Mays, *Water Resources Engineering*. Hoboken, NJ, USA: Wiley, 2010.
- [17] A. S. Zamzam, E. Dall'Anese, C. Zhao, J. A. Taylor, and N. D. Sidiropoulos, "Optimal water-power flow-problem: Formulation and distributed optimal solution," *IEEE Trans. Control Netw. Syst.*, vol. 6, no. 1, pp. 37–47, Mar. 2019.
- [18] K. Oikonomou and M. Parvania, "Optimal coordination of water distribution energy flexibility with power systems operation," *IEEE Trans. Power Syst.*, vol. 10, no. 1, pp. 1101–1110, Jan. 2019.
- [19] A. Stuhlmacher and J. L. Mathieu, "Chance-constrained water pumping to manage water and power demand uncertainty in distribution networks," *Proc. IEEE*, vol. 108, no. 9, pp. 1640–1655, Sep. 2020.
- [20] J. Najafi et al., "Resilience improvement planning of power-water distribution systems with multiple microgrids against hurricanes using clean strategies," *J. Cleaner Prod.*, vol. 223, pp. 109–126, 2019.
- [21] Q. Li, S. Yu, A. S. Al-Sumaiti, and K. Turitsyn, "Micro water-energy nexus: Optimal demand-side management and quasi-convex hull relaxation," *IEEE Trans. Control Netw. Syst.*, vol. 6, no. 4, pp. 1313–1322, Dec. 2019.
- [22] A. Belmondo Bianchi et al., "Distribution robust water-based demand side management in power transmission networks," *Sustain. Energy, Grids Netw.*, vol. 36, 2023, Art. no. 101232.
- [23] P. Lai, C. Courcoubetis, L. Duan, and S. Galelli, "Economic controls for smart water distribution networks undergoing supply failures," *IEEE Trans. Netw. Sci. Eng.*, vol. 8, no. 1, pp. 555–574, Jan.–Mar. 2021.
- [24] Y. Yao, C. Li, H.-M. Tai, and C. Shao, "Modeling of water transfer delay for optimal operation of integrated electrical-water system," *IEEE Trans. Power Syst.*, vol. 38, no. 1, pp. 331–348, Jan. 2023.
- [25] S. Zuloaga, P. Khatavkar, L. Mays, and V. Vittal, "Resilience of cyber-enabled electrical energy and water distribution systems considering infrastructural robustness under conditions of limited water and/or energy availability," *IEEE Trans. Eng. Manage.*, vol. 69, no. 3, pp. 639–655, Jun. 2022.
- [26] S. Zuloaga and V. Vittal, "Integrated electric power/water distribution system modeling and control under extreme mega drought scenarios," *IEEE Trans. Power Syst.*, vol. 36, no. 1, pp. 474–484, Jan. 2021.
- [27] M. Kitaura and M. Miyajima, "Damage to water supply pipelines," *Soils Foundations*, vol. 36, pp. 325–333, Jan. 1996.
- [28] Water Use for Electric Power Generation, *EPRI*, Palo Alto, CA: 2008, 1014026. Accessed: Apr. 18, 2025. [Online]. Available: <https://www.epri.com/research/products/1014026>
- [29] T. Ding et al., "Multi-stage distributionally robust stochastic dual dynamic programming to multi-period economic dispatch with virtual energy storage," *IEEE Trans. Sustain. Energy*, vol. 13, no. 1, pp. 146–158, Jan. 2022.
- [30] X. Yan and S. Liu, *Water Distribution and Drainage System*. Beijing, China: Architecture & Building Press, 2014.
- [31] T. M. Walsk et al., *Advanced Water Distribution Modeling and Management*. Exton, PA, USA: Bentley Inst. Press, 2007.
- [32] B. S. Rajkumar and A. Nielsen, "Water distribution in high rise buildings," 2010. Accessed: Apr. 18, 2025. [Online]. Available: <https://api.grundfos.com/literature/Grundfosliterature-6400727.pdf>
- [33] M. Movahednia, A. Kargarian, C. E. Ozdemir, and S. C. Hagen, "Power grid resilience enhancement via protecting electrical substations against flood hazards: A stochastic framework," *IEEE Trans. Ind. Informat.*, vol. 18, no. 3, pp. 2132–2143, Mar. 2022.
- [34] D. Sánchez-Muñoz et al., "Electrical grid risk assessment against flooding in barcelona and bristol cities," *Sustainability*, vol. 12, no. 4, Jan. 2020, Art. no. 1527.
- [35] S. Dunn et al., "Fragility curves for assessing the resilience of electricity networks constructed from an extensive fault database," *Natural Hazards Rev.*, vol. 19, no. 1, Feb. 2018, Art. no. 04017019.

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